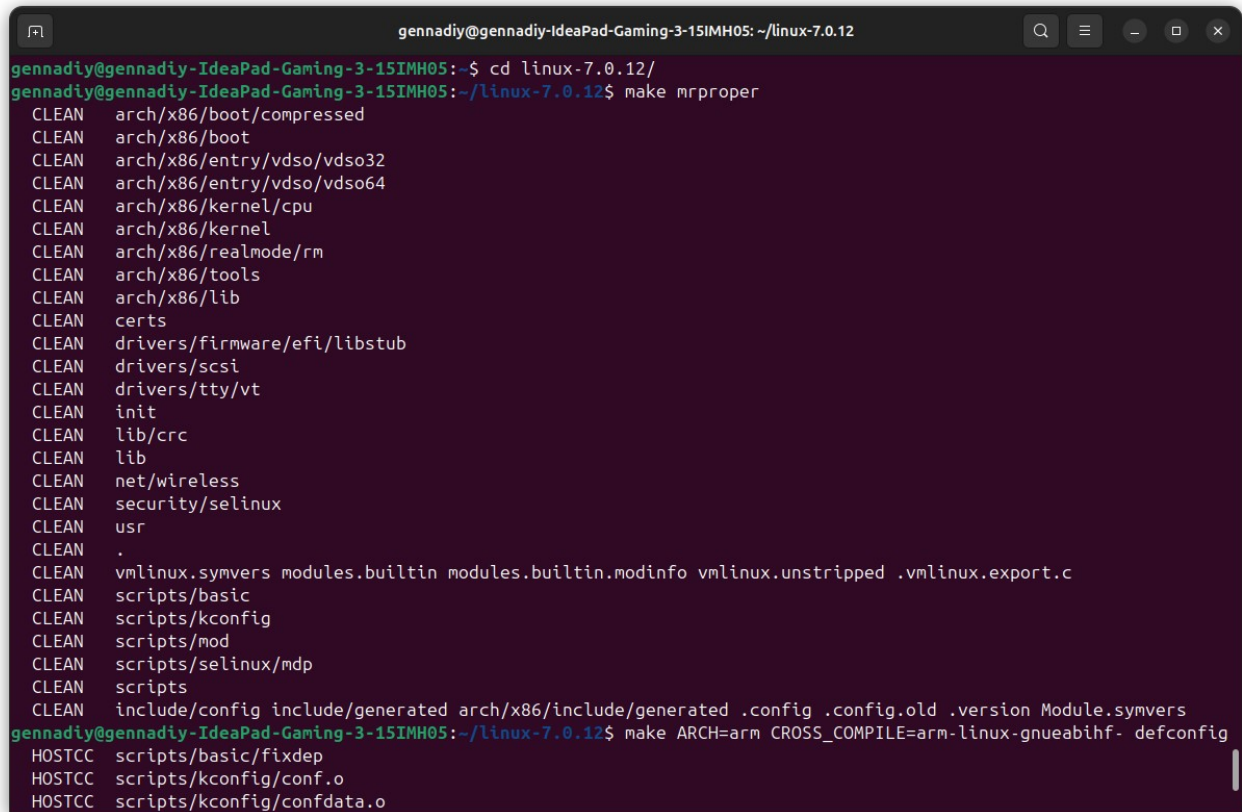


Задание 18 — Корневая файловая система (часть 1, получение ошибки)

- 1) устанавливаем QEMU и компилятор ARM командами `sudo apt update && sudo apt install qemu-system-arm` и `sudo apt update && sudo apt install gcc-arm-linux-gnueabi`
- 2) Выполняем `make mrproper` для очистки старых конфигураций



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ make mrproper
CLEAN arch/x86/boot/compressed
CLEAN arch/x86/boot
CLEAN arch/x86/entry/vdso/vdso32
CLEAN arch/x86/entry/vdso/vdso64
CLEAN arch/x86/kernel/cpu
CLEAN arch/x86/kernel
CLEAN arch/x86/realmode/rm
CLEAN arch/x86/tools
CLEAN arch/x86/lib
CLEAN certs
CLEAN drivers/firmware/efi/libstub
CLEAN drivers/scsi
CLEAN drivers/tty/vt
CLEAN init
CLEAN lib/crc
CLEAN lib
CLEAN net/wireless
CLEAN security/selinux
CLEAN usr
CLEAN .
CLEAN vmlinux.symvers modules.builtin modules.builtin.modinfo vmlinux.unstripped .vmlinux.export.c
CLEAN scripts/basic
CLEAN scripts/kconfig
CLEAN scripts/mod
CLEAN scripts/selinux/mdp
CLEAN scripts
CLEAN include/config include/generated arch/x86/include/generated .config .config.old .version Module.symvers
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
```

Рисунок 1 — удаление старых конфигураций

- 3) Конфигурация ядра командой `make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- defconfig`



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'multi_v7_defconfig'
#
# configuration written to .config
#
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$
```

Рисунок 2 — результат выполнения команды

- 4) Процесс компиляции исходного кода ядра linux 7.0.12 под ARM. На этом этапе также был скопирован образ ядра в рабочую папку

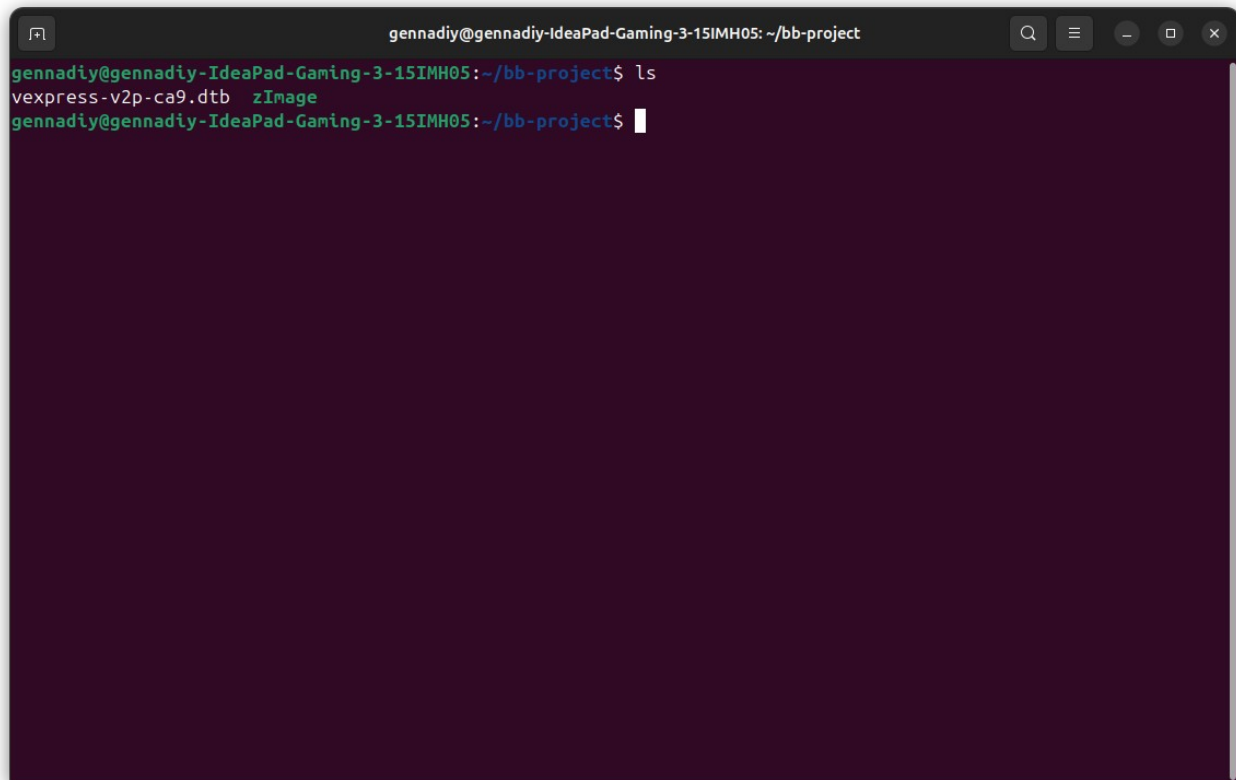
```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~  
GEN modules.builtin.modinfo  
GEN modules.builtin  
OBJCOPY arch/arm/boot/Image  
Kernel: arch/arm/boot/Image is ready  
LDS arch/arm/boot/compressed/vmlinux.lds  
AS arch/arm/boot/compressed/head.o  
GZIP arch/arm/boot/compressed/piggy_data  
CC arch/arm/boot/compressed/misc.o  
CC arch/arm/boot/compressed/decompress.o  
CC arch/arm/boot/compressed/string.o  
AS arch/arm/boot/compressed/hyp-stub.o  
CC arch/arm/boot/compressed/fdt_rw.o  
CC arch/arm/boot/compressed/fdt_ro.o  
CC arch/arm/boot/compressed/fdt_wip.o  
CC arch/arm/boot/compressed/fdt.o  
CC arch/arm/boot/compressed/atags_to_fdt.o  
CC arch/arm/boot/compressed/fdt_check_mem_start.o  
AS arch/arm/boot/compressed/lib1funcs.o  
AS arch/arm/boot/compressed/ashldi3.o  
AS arch/arm/boot/compressed/bswapsdi2.o  
AS arch/arm/boot/compressed/piggy.o  
LD arch/arm/boot/compressed/vmlinux  
OBJCOPY arch/arm/boot/zImage  
Kernel: arch/arm/boot/zImage is ready  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ cd ~  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ mkdir bb-project  
  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cp linux-7.0.12/arch/arm/boot/zImage bb-project/  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$
```

Рисунок 3 — результат компиляции

5) Подготовка аппаратной конфигурации DeviceTree командой `make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-hf- dtbs -j8`. Полученный файл `vexpress-v2p-ca9.dtb` скопирован в рабочую директорию командой `cp ~/linux-7.0.12/arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb ~/bb-project/`

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd linux-7.0.12/  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi-hf- dtbs -j8  
DTC arch/arm/boot/dts/actions/owl-s500-cubieboard6.dtb  
DTC arch/arm/boot/dts/airoha/en7523-evb.dtb  
DTC arch/arm/boot/dts/actions/owl-s500-guitar-bb-rev-b.dtb  
DTC arch/arm/boot/dts/amazon/alpine-db.dtb  
DTC arch/arm/boot/dts/arm/vexpress-v2p-ca5s.dtb  
DTC arch/arm/boot/dts/amlogic/meson8-minix-neo-x8.dtb  
DTC arch/arm/boot/dts/aspeed/aspeed-ast2500-evb.dtb  
DTC arch/arm/boot/dts/actions/owl-s500-labrador-base-m.dtb  
DTC arch/arm/boot/dts/allwinner/sun4i-a10-a1000.dtb  
DTC arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb  
DTC arch/arm/boot/dts/actions/owl-s500-roseapplepi.dtb  
DTC arch/arm/boot/dts/axis/artpec6-devboard.dtb  
DTC arch/arm/boot/dts/amlogic/meson8-fernseefee3.dtb  
DTC arch/arm/boot/dts/allwinner/sun4i-a10-ba10-tvbox.dtb  
DTC arch/arm/boot/dts/aspeed/aspeed-ast2600-evb-a1.dtb  
DTC arch/arm/boot/dts/broadcom/bcm2835-rpi-b.dtb  
DTC arch/arm/boot/dts/arm/vexpress-v2p-ca15-tc1.dtb  
DTC arch/arm/boot/dts/actions/owl-s500-sparkly.dtb  
DTC arch/arm/boot/dts/arm/vexpress-v2p-ca15-a7.dtb  
DTC arch/arm/boot/dts/broadcom/bcm2835-rpi-a.dtb  
DTC arch/arm/boot/dts/amlogic/meson8b-ec100.dtb  
DTC arch/arm/boot/dts/allwinner/sun4i-a10-chuwi-v7-cw0825.dtb  
DTC arch/arm/boot/dts/allwinner/sun4i-a10-cubieboard.dtb  
DTC arch/arm/boot/dts/aspeed/aspeed-ast2600-evb.dtb  
DTC arch/arm/boot/dts/amlogic/meson8b-mxq.dtb  
DTC arch/arm/boot/dts/broadcom/bcm2835-rpi-b-rev2.dtb  
DTC arch/arm/boot/dts/aspeed/aspeed-bmc-amd-daytonax.dtb
```

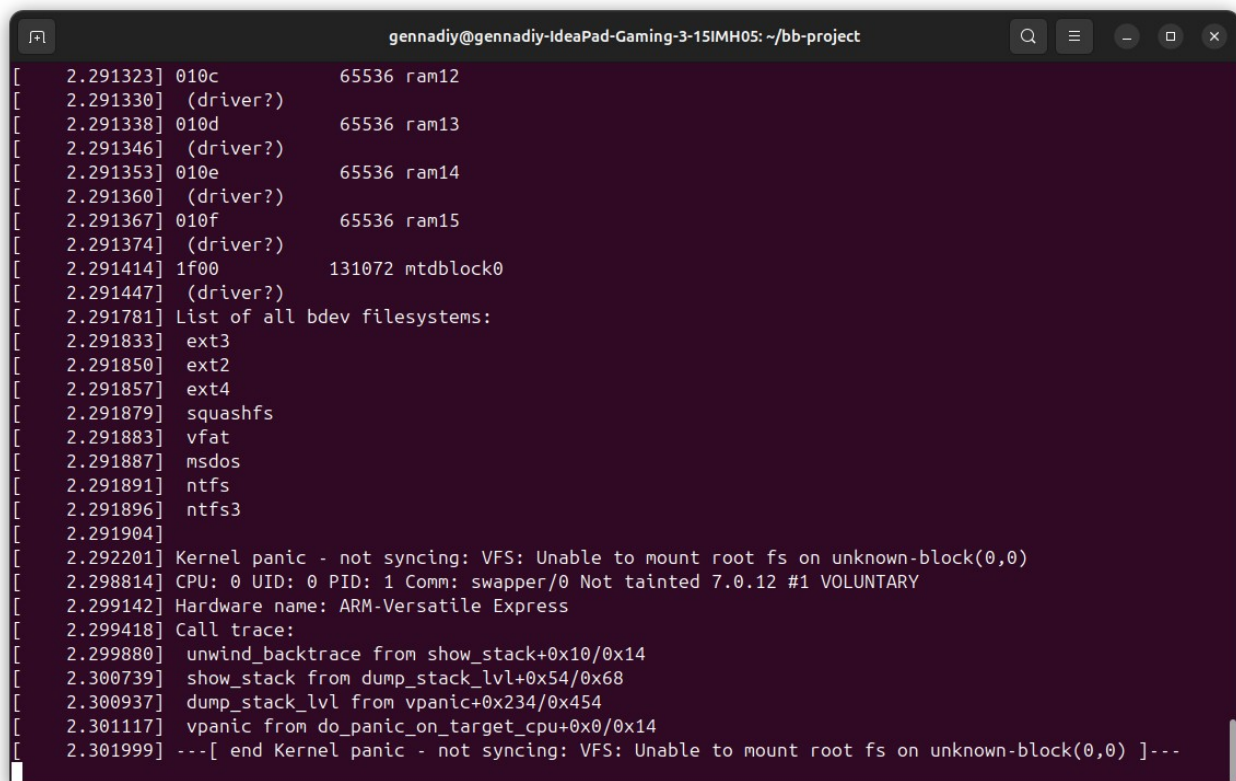
Рисунок 4 — компиляция файлов DeviceTree



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ ls
vexpress-v2p-ca9.dtb  zImage
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$
```

Рисунок 5 — рабочая директория

6) Для эмуляции ядра внутри QEMU выполняем команду `QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic`



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
[ 2.291323] 010c 65536 ram12
[ 2.291330] (driver?)
[ 2.291338] 010d 65536 ram13
[ 2.291346] (driver?)
[ 2.291353] 010e 65536 ram14
[ 2.291360] (driver?)
[ 2.291367] 010f 65536 ram15
[ 2.291374] (driver?)
[ 2.291414] 1f00 131072 mtdblock0
[ 2.291447] (driver?)
[ 2.291781] List of all bdev filesystems:
[ 2.291833] ext3
[ 2.291850] ext2
[ 2.291857] ext4
[ 2.291879] squashfs
[ 2.291883] vfat
[ 2.291887] msdos
[ 2.291891] ntfs
[ 2.291896] ntfs3
[ 2.291904]
[ 2.292201] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
[ 2.298814] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Not tainted 7.0.12 #1 VOLUNTARY
[ 2.299142] Hardware name: ARM-Versatile Express
[ 2.299418] Call trace:
[ 2.299880] unwind_backtrace from show_stack+0x10/0x14
[ 2.300739] show_stack from dump_stack_lvl+0x54/0x68
[ 2.300937] dump_stack_lvl from vpanic+0x234/0x454
[ 2.301117] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 2.301999] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---
```

Рисунок 6 — Ошибка kernel panic при запуске ядра без корневой файловой системы